

STREPSILS MAX PRO DIRECT SPRAY

Flurbiprofen 8.75mg

Product Description:

Oromucosal spray, solution. Clear, colourless to slightly yellow solution.

The active ingredient is flurbiprofen.

The other ingredients are: Betadex, Disodium phosphate dodecahydrate, Citric acid monohydrate, Methyl parahydroxybenzoate (E218), Propyl parahydroxybenzoate (E216), Sodium hydroxide, Mint flavour, Cherry flavour, N,2,3-Trimethyl-2-isopropylbutanamide, Saccharin sodium, Hydroxypropyl betadex and Purified water.

Pharmacodynamics

Flurbiprofen is a propionic acid derivative NSAID which acts through inhibition of prostaglandin synthesis. In humans flurbiprofen has potent analgesic, antipyretic and anti-inflammatory properties and the 8.75 mg dose dissolved in artificial saliva has been shown to reduce prostaglandin synthesis in cultured human respiratory cells. According to studies using the whole blood assay, flurbiprofen is a mixed COX-1/COX-2 inhibitor with some selectivity towards COX-1. Pre-clinical studies suggest that the R (-) enantiomer of flurbiprofen and related NSAIDs may act on the central nervous system; the suggested mechanism is by inhibition of induced COX-2 at the level of the spinal cord.

A single dose of flurbiprofen 8.75 mg delivered locally to the throat as three sprays has been demonstrated to relieve sore throat, including swollen and inflamed sore throats through a significant change in the severity of throat soreness area under the change (AUC) from baseline curve (mean difference (standard deviation)) for active treatment versus placebo from 0 to 2 hours (-1.82 (1.35) vs -1.13 (1.14)), 0 to 3 hours (-2.01 (1.405) vs -1.31 (1.233)) and 0 to 6 hours (-2.14 (1.551) vs -1.50 (1.385)). Significant differences in the AUC from baseline curve from 0 – 6 hours compared to placebo were also seen for other qualities of sore throat including pain intensity (-22.50 (17.894) vs -15.64 (16.413)), difficulty swallowing (-22.50 (18.260) vs -16.01 (15.451)), swollen throat (-20.97 (18.897) vs -13.80 (15.565)) and sore throat pain relief (3.24 (1.456) vs 2.47 (1.248)). The change from baseline at individual time points across the different qualities of sore throat demonstrated significance starting from 5 minutes and lasting for up to 6 hours.

For those patients taking antibiotics for Strep infection, there was statistically significant greater relief of sore throat pain intensity for flurbiprofen 8.75 mg lozenge from 7 hours and onwards after antibiotics were taken. The analgesic effect of flurbiprofen 8.75 mg lozenge was not reduced by the administration of antibiotics to treat patients with streptococcal sore throat.

Multiple-dose efficacy over 3 days has also been demonstrated. The mint-flavoured spray is easy and convenient to use and reaches the sore area of the throat, improving the ability to talk and use the voice whilst providing a soothing and coating effect for the throat.

Paediatric Population

No specific studies in children have been undertaken with Strepsils Max Pro Direct Spray 8.75mg per dose. Efficacy and safety studies on flurbiprofen 8.75 mg lozenges have included children aged 12 – 17 years, although small sample size means that no statistical conclusions can be drawn.

Pharmacokinetics

Absorption

A single dose of flurbiprofen 8.75 mg is delivered directly to the throat as three sprays and the flurbiprofen is readily absorbed, with detection in the blood between 2 and 5 minutes and plasma concentrations peaking at 30 minutes after administration, but remaining at a mean low level of 1.6 µg/mL which is approximately 4 times lower than a 50 mg tablet dose. Strepsils Max Pro Direct Spray 8.75mg per dose demonstrates bioequivalence to flurbiprofen 8.75 mg lozenge. Absorption of flurbiprofen can occur from the buccal cavity by passive diffusion. Rate of absorption is dependent on pharmaceutical form with peak concentrations achieved more rapidly than, but of similar magnitude to, those achieved after an equivalent swallowed dose.

Distribution

Flurbiprofen is rapidly distributed throughout the body and is extensively bound to plasma proteins.

Metabolism / Excretion

Flurbiprofen is mainly metabolised by hydroxylation and excreted via the kidneys. It has an elimination half-life of 3 to 6 hours. Flurbiprofen is excreted in very small amounts in human milk (less than 0.05 µg/ml). Approximately 20-25% of a flurbiprofen oral dose is excreted unchanged.

Special Groups

No difference in pharmacokinetic parameters between elderly and young adult volunteers has been reported following oral administration of flurbiprofen tablets. No pharmacokinetic data have been generated in children below 12 years of age following administration of Flurbiprofen 8.75 mg however administration of both flurbiprofen syrup and suppository formulations indicate no significant differences in pharmacokinetic parameters compared with adults.

Indication

Strepsils Max Pro Direct Spray 8.75mg per dose is indicated for the short term symptomatic relief of acute sore throat in adults.

Recommended Dosage

Adults aged 18 years and over:

One dose (3 sprays) administered to the back of the throat every 3-6 hours as required, up to a maximum of 5 doses in a 24 hour period.

Do not inhale whilst spraying.

It is recommended that this product should be used for a maximum of three days.

Before first use, activate the pump by pointing the nozzle away from you and spraying a minimum of four times until a fine, consistent mist is produced. The pump is then primed and ready for use.

Between each dose point the nozzle away from you and spray a minimum of once ensuring a fine, consistent mist is produced. Always ensure a fine consistent mist is produced before dosing the product.

Paediatric population

The safety and efficacy of Strepsils Max Pro Direct Spray 8.75mg per dose in children or adolescents under 18 years has not been established.

Elderly patients

A general dose recommendation cannot be given, since to date clinical experience is limited.

The elderly are at increased risk of the serious consequences of adverse reactions.

The lowest effective dose should be administered for the shortest duration necessary to control symptoms (see section Warnings and Precautions).

Route of Administration

For oromucosal administration and short-term use only.

Contraindications

- Hypersensitivity to flurbiprofen or to any of the excipients
- Patients who have previously shown hypersensitivity reactions (e.g asthma, bronchospasm, rhinitis, angioedema or urticaria) in response to acetylsalicylic acid or other NSAIDs.
- Active, or history of recurrent peptic ulcer/haemorrhage (two or more distinct episodes of proven ulceration) and intestinal ulceration.
- History of gastrointestinal bleeding or perforation, severe colitis, haemorrhagic or haematopoietic disorders related to previous NSAID therapy.
- Last trimester of pregnancy (See section Statement on usage during pregnancy and lactation)
- Severe heart failure, severe renal failure or severe hepatic failure (see section Warnings and Precautions).
- Children and adolescents below 18 years.

Warnings and Precautions

Undesirable effects may be minimised by using the lowest effective dose for the shortest duration necessary to control symptoms.

Infections

Since in isolated cases an exacerbation of infective inflammations (e.g. development of necrotising fasciitis) has been described in temporal association with the use of systemic NSAIDs as a class, the patient is advised to consult a physician immediately if signs of a bacterial infection occur or worsen during the flurbiprofen spray therapy. It should be considered whether initiation of an anti-infective antibiotic therapy is indicated.

In cases of purulent bacterial pharyngitis/tonsillitis, the patient is advised to consult a physician as the treatment needs to be re-evaluated.

Treatment should be administered for three days maximum.

If the symptoms get worse or if new symptoms occur, the treatment should be re-evaluated.

If mouth irritation occurs, flurbiprofen treatment should be withdrawn.

Elderly population

The elderly have an increased frequency of adverse reactions to NSAIDs especially gastrointestinal bleeding and perforation which may be fatal.

Respiratory

Bronchospasm may be precipitated in patients suffering from or with a previous history of bronchial asthma or allergic disease. Flurbiprofen spray should be used with caution in these patients.

Other NSAIDs

The use of flurbiprofen spray with concomitant NSAIDs including cyclooxygenase-2 selective inhibitors should be avoided (see section Interactions with Other Medicaments).

Systemic lupus erythematosus and mixed connective tissue disease

Patients with systemic lupus erythematosus and mixed connective tissue disease may have an increased risk of aseptic meningitis (see section Undesirable Effects), however this effect is not usually seen with short term limited use products such as flurbiprofen spray.

Cardiovascular, Renal and Hepatic Impairment

NSAIDs have been reported to cause nephrotoxicity in various forms including interstitial nephritis, nephrotic syndrome and renal failure. The administration of an NSAID may cause a dose dependent reduction in prostaglandin formation and precipitate renal failure. Patients at greatest risk of this reaction are those with impaired renal function, cardiac impairment, liver dysfunction, those taking diuretics and the elderly, however, this effect is not usually seen with short term, limited use products such as flurbiprofen spray.

Hepatic

Mild to moderate hepatic dysfunction (see sections Contraindications and Undesirable Effects).

Cardiovascular and cerebrovascular effects

Caution (discussion with doctor or pharmacist) is required prior to starting treatment in patients with a history of hypertension and/or heart failure as fluid retention, hypertension and oedema have been reported in association with NSAID therapy.

Clinical trial and epidemiological data suggest that use of some NSAIDs, (particularly at high doses and in long term treatment) may be associated with a small increased risk of arterial thrombotic events (for example myocardial infarction or stroke). There are insufficient data to exclude such a risk for flurbiprofen when given at a daily dose of no more than 5 doses (3 sprays per dose). All NSAIDs should not be used perioperatively in patients who have recently undergone coronary artery bypass graft (CABG) surgery and revascularisation procedures.

Nervous System effects

Analgesic induced headache - In the event of prolonged use of analgesics or use beyond the regulations headache may occur, which must not be treated with increased doses of the medicinal product.

Gastrointestinal

NSAIDs should be given with care to patients with a history of gastrointestinal disease (ulcerative colitis, Crohn's disease) as these conditions may be exacerbated (see section Undesirable Effects).

Gastrointestinal bleeding, ulceration or perforation, which can be fatal, has been reported with all NSAIDs at any time during treatment, with or without warning symptoms or a previous history of serious GI events.

The risk of GI bleeding, ulceration or perforation is higher with increasing NSAID doses, in patients with a history of ulcer, particularly if complicated with haemorrhage or perforation (see section Contraindications), and in the elderly, however this effect is not usually seen with short term limited use products such as flurbiprofen spray. Patients with a history of GI toxicity, particularly when elderly, should report any unusual abdominal symptoms (especially GI bleeding) to their healthcare professional.

Caution should be advised in patients receiving concomitant medications which could increase the risk of ulceration or bleeding, such as oral corticosteroids, anticoagulants such as warfarin, selective serotonin-reuptake inhibitors or anti-platelet agents such as acetylsalicylic acid (see section Interactions with Other Medicaments).

If GI bleeding or ulceration occurs in patients receiving flurbiprofen, the treatment should be withdrawn.

Haematological effects

Flurbiprofen, like other NSAIDs, may inhibit platelet aggregation and prolong bleeding time. Flurbiprofen spray should be used with caution in patients with a potential for abnormal bleeding.

Dermatological

Serious skin reactions, some of them fatal, including exfoliative dermatitis, Stevens-Johnson syndrome and toxic epidermal necrolysis, have been reported very rarely in association with the use of NSAIDs (see section Undesirable Effects). Flurbiprofen spray should be discontinued at the first appearance of skin rash, mucosal lesions, or any other sign of hypersensitivity.

This product contains methyl parahydroxybenzoate and propyl parahydroxybenzoate which may cause allergic reactions (possibly delayed).

Risk of GI Ulceration, Bleeding and Perforation with NSAID

Serious GI toxicity such as bleeding, ulceration and perforation can occur at any time, with or without warning symptoms, in patients treated with NSAID therapy. Although minor upper GI problems (e.g. dyspepsia) are common, usually developing early in therapy, prescribers should remain alert for ulceration and bleeding in patients treated with NSAIDs even in the absence of previous GI tract symptoms.

Studies to date have not identified any subset of patients not at risk of developing peptic ulceration and bleeding. Patients with prior history of serious GI events and other risk factors associated with peptic ulcer disease (e.g. alcoholism, smoking, and corticosteroid therapy) are at increased risk. Elderly or debilitated patients seem to tolerate ulceration or bleeding less than other individuals and account for most spontaneous reports for fatal GI events.

Interactions with Other Medicaments

Flurbiprofen should be avoided in combination with:	
Other NSAIDs including cyclooxygenase-2 selective inhibitors:	Avoid concomitant use of two or more NSAIDs as this may increase the risk of adverse effects (esp. gastrointestinal adverse events such as ulcers and bleeding), (see section Warnings and Precautions).
Acetylsalicylic acid (low dose)	Unless low-dose aspirin (not above 75mg daily) has been advised by a doctor, as this may increase the risk of adverse reactions (see section Warnings and Precautions).

Flurbiprofen should be used with caution in combination with:	
Anticoagulants	NSAIDs may enhance the effects of anti-coagulants, such as warfarin (see section Warnings and Precautions).
Anti-platelet Agents	Increased risk of gastrointestinal ulceration or bleeding (see section Warnings and Precautions).
Antihypertensive drugs (Diuretics, ACE inhibitors, angiotensin-II-antagonists):	NSAIDs may reduce the effect of diuretics and other antihypertensive drugs may enhance nephrotoxicity caused by inhibition of cyclooxygenase, especially in patients with compromised renal function
Alcohol	May increase the risk of adverse reactions, especially of bleeding in the gastrointestinal tract
Cardiac glycosides:	NSAIDs may exacerbate cardiac failure, reduce GFR and increase plasma glycoside levels - adequate control and, if necessary, dose adjustment is recommended
Ciclosporin:	Increased risk of nephrotoxicity.
Corticosteroids:	Increased risk of gastrointestinal ulceration or bleeding (see section

	Warnings and Precautions)
Lithium:	May increase serum levels of lithium – adequate control and, if necessary, dose adjustment is recommended
Methotrexate:	The administration of NSAIDs within 24 hours before or after administration of methotrexate may lead to elevated concentrations of methotrexate and an increase in its toxic effect.
Mifepristone:	NSAIDs should not be used for 8 – 12 days after mifepristone administration as NSAIDs can reduce the effect of mifepristone.
Oral antidiabetics	Alteration of blood glucose levels reported (increased check rate recommended)
Phenytoin	May increase serum levels of phenytoin – adequate control and, if necessary, dose adjustment is recommended
Potassium sparing diuretics	Concomitant use may cause hyperkalaemia
Probenecid Sulfapyrazone	Medicinal products that contain probenecid or sulfapyrazone may delay the excretion of flurbiprofen.
Quinolone antibiotics	Animal data indicate that NSAIDs can increase the risk of convulsions associated with quinolone antibiotics. Patients taking NSAIDs and quinolones may have an increased risk of developing convulsions.
Selective serotonin reuptake inhibitors (SSRI's)	Increased risk of gastrointestinal ulceration or bleeding (see section Warnings and Precautions).
Tacrolimus:	Possible increased risk of nephrotoxicity when NSAIDs are given with tacrolimus.
Zidovudine:	Increased risk of haematological toxicity when NSAIDs are given with zidovudine.

No studies so far have revealed any interactions between flurbiprofen and tolbutamide or antacids.

Paediatric population

No additional information available.

Statement on usage during pregnancy and lactation

Pregnancy

Inhibition of prostaglandin synthesis may adversely affect the pregnancy and/or the embryo/foetal development. Data from epidemiological studies suggest an increased risk of miscarriage and of cardiac malformation and gastroschisis after use of a prostaglandin synthesis inhibitor in early pregnancy. The absolute risk for cardiovascular malformation was increased from less than 1%, up to approximately 1.5 %. The risk is believed to increase with dose and duration of therapy. In animals, administration of a prostaglandin synthesis inhibitor has been shown to result in increased pre- and post-implantation loss and embryo-foetal lethality. In addition, increased incidences of various malformations, including cardiovascular, have been reported in animals given a prostaglandin synthesis inhibitor during the organogenetic period. During the first and second trimester of pregnancy, flurbiprofen should not be given.

During the third trimester of pregnancy, all prostaglandin synthesis inhibitors may expose

- the foetus to:
 - cardiopulmonary toxicity (with premature closure of the ductus arteriosus and pulmonary hypertension);
 - renal dysfunction, which may progress to renal failure with oligo-hydroamniosis;
 - the mother and the neonate, at the end of pregnancy, to:
 - possible prolongation of bleeding time, an anti-aggregating effect which may occur even at very low doses.
 - inhibition of uterine contractions resulting in delayed or prolonged labour.
- Consequently, flurbiprofen is contraindicated during the third trimester of pregnancy (see section Contraindications).

Breastfeeding

In limited studies, flurbiprofen appears in the breast milk in very low concentration and is unlikely to affect the breast-fed infant adversely. However, because of possible adverse effects of NSAIDs on breast-fed infants, flurbiprofen spray is not recommended for use in nursing mothers.

Fertility

There is some evidence that drugs which inhibit cyclo-oxygenase/ prostaglandin synthesis may cause impairment of female fertility by an effect on ovulation. This is reversible on withdrawal of treatment.

Undesirable Effects

Hypersensitivity reactions to NSAIDs have been reported and these may consist of:

- (a) Non-specific allergic reactions and anaphylaxis
- (b) Respiratory tract reactivity, e.g. asthma, aggravated asthma, bronchospasm, dyspnoea
- (c) Various skin reactions, e.g. pruritus, urticaria, angioedema and more rarely exfoliative and bullous dermatoses (including epidermal necrolysis and erythema multiforme).

Oedema, hypertension and cardiac failure have been reported in association with NSAID treatment. There is insufficient data to exclude such a risk for flurbiprofen oromucosal spray, solution.

The following list of adverse effects relates to those experienced with flurbiprofen at OTC doses for short-term use. (Very common ($\geq 1/10$), Common ($\geq 1/100$ to $< 1/10$), Uncommon ($\geq 1/1000$ to $< 1/100$), Rare ($\geq 1/10000$ to $< 1/1000$), Very rare ($< 1/10000$), not known (cannot be estimated from the available data))

Blood and lymphatic system disorders:

Not known: anaemia, thrombocytopenia.

Cardiovascular and cerebrovascular disorders:

Not known: Oedema, hypertension, cardiac failure

Nervous System disorders:

Common: dizziness, headache, paresthesia

Uncommon: somnolence

Respiratory, thoracic and mediastinal disorders:

Common: throat irritation

Uncommon: exacerbation of asthma and bronchospasm, dyspnoea, wheezing, oropharyngeal blistering, pharyngeal hypoaesthesia.

Gastrointestinal disorders:

Common: diarrhoea, mouth ulceration, nausea, oral pain, paraesthesia oral, oropharyngeal pain, oral discomfort (warm or burning feeling or tingling of the mouth).

Uncommon: abdominal distension, abdominal pain, constipation, dry mouth, dyspepsia, flatulence, glossodynia, dysgeusia, oral dysaesthesia, vomiting

Skin and subcutaneous tissue disorders:

Uncommon: various skin rashes, pruritus.

Not known: severe forms of skin reaction such as bullous reactions, including Stevens-Johnson syndrome and toxic epidermal necrolysis.

General disorders and administration site conditions:

Uncommon: pyrexia, pain

Immune System disorders:

Rare: anaphylactic reaction

Psychiatric disorders:

Uncommon: insomnia

Hepatobiliary disorders:

Not known: hepatitis

Overdose and Treatment

Symptoms

Most patients who have ingested clinically important amounts of NSAIDs will develop no more than nausea, vomiting, epigastric pain, or more rarely diarrhoea. Tinnitus, headache, and gastrointestinal bleeding are also possible. In more serious poisoning with NSAIDs, toxicity is seen in the central nervous system, manifesting as drowsiness, occasionally excitation, blurred vision and disorientation or coma. Occasionally patients develop convulsions. In serious poisoning with NSAIDs metabolic acidosis may occur and the prothrombin time/INR may be prolonged, probably due to interference with the actions of circulating clotting factors. Acute renal failure and liver damage may occur. Exacerbation of asthma is possible in asthmatics.

Management

Management should be symptomatic and supportive and include the maintenance of a clear airway and monitoring of cardiac and vital signs until stable. Consider oral administration of activated charcoal and if necessary correction of serum electrolytes if the patient presents within one hour of ingestion or a potentially toxic amount. If frequent or prolonged, convulsions should be treated with intravenous diazepam or lorazepam. Give bronchodilators for asthma. There is no specific antidote to flurbiprofen.

Storage Conditions

Do not store above 30°C. Do not refrigerate or freeze. Do not use this medicine for more than 6 months after the first use.

Packaging available

Available in 15ml spray bottle

Name and address of product registration holder:

RB (Health) Malaysia Sdn Bhd (1297081-V)

Level 17, Menara One Sentrum,

No. 201, Jalan Tun Sambanthan,

50470 Kuala Lumpur, Malaysia.

Careline number: 1800-888-889

Reckitt Benckiser (Singapore) Pte. Ltd.

12 Marina Boulevard #19 – 01, Marina Bay Financial Centre,
Tower 3, Singapore 018982.

Careline number: 1-800-875-0770

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